Junaid Rehman

Neuromorphic Research Engineer & Co-founder | Islamabad, Pakistan

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Summary

Trained as an electronics engineer and now working as a Neuromorphic Research Engineer, I focus on bridging the performance gap between artificial neural networks and physical hardware. I specialize in developing 'hardware-aware' solutions that translate brain-inspired computational efficiency into silicon. My work centers on the co-evolution of algorithms and hardware to ensure high-performance intelligence that is both constrained by, and optimized for, the physics of the chip. I aim to build systems where the hardware doesn't just execute the AI, but actively enables its efficiency—achieving significant energy improvements through architectural synergy.

Education

Pakistan Institute of Engineering and Applied Sciences (PIEAS) — Islamabad, Pakistan

BS Electrical Engineering (Electronics)

Relevant Coursework: VLSI Design, Integrated Electronics, Digital Signal Processing | Merit Scholarship Recipient

Selected Publications

Modality-Dependent Memory Mechanisms in Cross-Modal Neuromorphic Computing — 2026

arXiv Preprint (Under Review)

- First comprehensive cross-modal ablation study evaluating Hopfield networks, HGRNs, and supervised contrastive learning (SCL) across visual (N-MNIST) and auditory (SHD) neuromorphic datasets.
- Revealed striking modality-dependent patterns: Hopfield networks excel at visual tasks (**97.68%**) but suffer a **21.53 point drop** on auditory tasks (76.15%), while standalone SCL offers robust cross-modal averages (89.44%).
- Proposed **Full Hybrid (M5)** architecture achieves peak cross-modal average of **89.99%** (97.72% visual; 82.25% auditory); optimized HGRN (M4) achieves **719.0**× energy-efficiency improvement over dense ANNs with **99.3%** spike sparsity.

Research & Development

Project Phasor — Remote

Neuromorphic Research Engineer — *2025 – Present*

- Architecting hardware-aware SNNs that translate cortical column dynamics into silicon-ready models; co-designing algorithms with constraints of analog/mixed-signal hardware.
- Engineering engram-based memory layers using PyTorch/snntorch to achieve sub-microwatt inference targets for edge deployment; collaborating with neuroscientists on biologically plausible learning rules.

GSME — Islamabad, Pakistan

Analog/Mixed-Signal Design Research Assistant — *2022 – 2023*

- Tapeout-ready design of **65dB gain operational amplifier** in TSMC 65nm & 130nm; full GDSII layout with advanced matching techniques, DRC/LVS clean.
- Designed **200MHz PLL** with wide tuning range for high-speed ADC/DAC clocking; post-layout simulation including parasitic extraction (PEX).
- Developed delay-free insensitive switched-capacitor integrator; designed non-overlapping clock generator (40ns period, 5ns dead-time) in TSMC 130nm.
- Tools: Cadence Virtuoso, Spectre, Xschem, Netgen, Calibre, TSMC/SKY130/GF180 PDKs.

Entrepreneurship

Two Apps — Dubai, UAE (Remote)

Co-founder — *2025 – Present*

- Co-founded UAE-based AI automation agency delivering **Agentic AI workflows** and white-label AI implementation for businesses and software houses globally (Eastern Europe, South America, Australia).
- Architected end-to-end AI automation systems for finance operations, AML/KYC compliance, and back-office workflows using Claude, GPT-4, and agentic frameworks; reduced client manual processing by 40–60%.
- Established white-label delivery partnerships enabling agencies to deliver AI projects without expanding internal ML teams; implemented structured delivery methodology (Audit → Pilot → Stabilize → Scale).
- Led technical product development using Flutter, AWS, and modern LLM orchestration (LangChain, Ollama), managing full-stack delivery from concept to production.
- Managed client relationships and compliance-aware AI deployments for fintech and regulated industries, ensuring human-in-the-loop oversight and production reliability.

Industry Experience

IdrakAI Islamabad, Pakistan
AI Engineer *2022 – 2024*

- Deployed production OCR pipelines processing **10,000+ daily documents** for banking verification; reduced latency **to 0.1s** via chunking and INT8 quantization.
- Fine-tuned **TinyLlama & Llama 3** using ORPO (Odds Ratio Preference Optimization) for domain-specific tasks; deployed on-device via Ollama + LangChain.
- Engineered a detection system using Computer Vision; built Call Analytics dashboard with Speaker Diarization (PyAnnote) for financial compliance.
- Optimized Whisper ASR and StyleTTS pipelines for real-time banking applications; established MLOps workflows for model versioning and A/B testing.

Technical Skills

Neuromorphic:	SNNs, LIF neurons, Event-based Vision, snntorch, Norse, Brain-inspired algorithms
LLM/GenAI:	PyTorch, Transformers, ORPO, RLHF, Quantization (GPTQ/AWQ), Ollama, LangChain, Agentic AI
Digital Design:	Verilog, SystemVerilog, RTL Design, Synthesis, STA, FPGA prototyping
Analog/RF IC:	Cadence Virtuoso, Spectre, GDSII Layout, PLLs, Op-Amps, SC Circuits, PEX
Process Tech:	TSMC 65nm, TSMC 130nm, GlobalFoundries 180nm, SkyWater 130nm (SKY130)
Full-Stack:	Python, C++, Flutter, AWS, Docker, Git, Linux, REST APIs

Selected Achievements

PIEAS University Scholarship – Merit-based award for outstanding academic performance.
National Level Rugby Player – Playing in the DIV 1 league in Pakistan.
Languages: English (C1 Professional Proficiency) | Urdu (Native) | Arabic (Basic)

References available upon request.